

Fast Complete BN254 Scalar Multiplication

Optimal recoding and complete RCB addition for ArgoMAC

Kriterion research submission

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Abstract

The ArgoMAC baseline uses an incomplete point-addition law and 92 digits in base $2 - \omega$. This report gives a complete 91-digit construction for BN254 G1. An exact Eisenstein-norm proof establishes the digit bound. A second Lean proof establishes complete Renes–Costello–Batina addition. It covers ordinary addition, doubling, and inverse addition. The construction uses the minimum adaptor count in the fixed bidegree $(2, 2)$ Bosma–Lenstra family. Lean 4.24 checks these results without a new axiom or compiler-trusted evaluation. It also checks the 508-bit Lamport label order. The matching Rust code uses the same law and garbles independent rows with Rayon.

1 Contribution

Let r be the BN254 scalar modulus. Let ω be the nontrivial cube root used by the G1 endomorphism. Set $\beta = 2 - \omega$. The digit alphabet is

$$D = \{0, \pm 1, \pm \omega, \pm \omega^2\}.$$

The norm of β is seven.

The baseline uses 92 fixed positions. The new construction uses 91 fixed positions for every scalar. It keeps the same alphabet and input-label shape. It also replaces the exceptional affine row with one complete homogeneous row. The complete row increases the per-row adaptor count from nine to 13. Shared garbling passes and parallel row execution offset this extra work.

2 Four-corner GLV recoding

Use the Eisenstein norm

$$Q(a, b) = a^2 - ab + b^2.$$

The relevant BN254 constants are

$$\begin{aligned} A &= 147946756881789319000765030803803410728, \\ d &= 9931322734385697763, \\ C &= A + d, \\ r &= A^2 + Ad + d^2. \end{aligned}$$

The two GLV kernel vectors are

$$v_1 = (-A, d), \quad v_2 = (-d, -C).$$

Arkworks divides the two negative quotient numerators toward zero. Thus, its GLV state is the floor representative z_0 used by the proof. The selector checks

$$z_0, \quad z_0 + v_1, \quad z_0 + v_2, \quad z_0 + v_1 + v_2.$$

Each shift is in the reconstruction kernel. Thus, every candidate represents the same scalar.

For quotient remainders $u, v \in [0, r)$, scaling maps these four states to the four corners of one Eisenstein cell. The two triangle lemmas show that one corner satisfies

$$3Q(a, b) \leq r.$$

This is the exact covering-radius step.

3 Exact 91-round bound

One recurrence round selects a unit digit and divides by β . For a state with $Q(a, b) \leq m$, the selected digit gives

$$Q(a - u_d, b - v_d) \leq m + 1 + \lfloor \sqrt{4m} \rfloor.$$

Exact division gives

$$Q(a - u_d, b - v_d) = 7Q(a', b').$$

Define the inverse safe bound

$$\begin{aligned} B(0) &= 2, \\ B(n+1) &= 7B(n) + 1 - \left\lfloor \sqrt{4(7B(n) + 1)} \right\rfloor. \end{aligned}$$

Lean proves the generic induction rule

$$Q(a, b) \leq B(n) \implies \text{after}(n+1, (a, b)) = (0, 0).$$

It also checks

$$\begin{aligned} B(90) &= 8072890822786049911192470628436961768360750887255955910449276353626847573707, \\ r &\leq 3B(90). \end{aligned}$$

The numeric proof uses ten-round kernel-checked checkpoints. It does not use `native_decide`.

The main Lean results are:

- `shiftedInitialCorrect`;
- `existsCandidateNormBound`;
- `fourCandidatesSignedI128`;
- `existsCorrectShiftedInitialTerminates91`;
- `shortInitialTerminates91`;
- `shortInitialCorrect`.

The axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`.

4 Optimality

Lean checks

$$7^{90} < r < 7^{91}.$$

A fixed 90-position string over seven symbols has at most 7^{90} values. It cannot represent all r scalars. Thus, 91 is optimal for this fixed alphabet.

5 Complete homogeneous addition

Let the hidden offset be $K = (A : B : C)$. Let (u, v) be the selected endomorphism image of the evaluator input. Use homogeneous coordinates for

$$Y^2Z = X^3 + 3Z^3.$$

An affine point is $(x : y : 1)$. The group identity is $(0 : 1 : 0)$.

RCB Algorithm 7 gives these three polynomials:

$$\begin{aligned} X &= -9AB - 18BCu - 18ACv + (B^2 - 9C^2)uv + ABv^2, \\ Y &= -81C^2 + 27A^2u + 27ACu^2 + B^2v^2, \\ Z &= 9BC + (B^2 + 9C^2)v + BCv^2 + 3A^2uv + 3ABu^2. \end{aligned}$$

One fresh nonzero value ρ multiplies all three coordinates. This common factor preserves the represented projective point. The zero digit returns $\rho(A, B, C)$.

The decoder keeps Z until the final projection. For $Z \neq 0$, it validates $(X/Z, Y/Z)$ and returns that point. For $Z = 0$, it accepts only $X = 0$ and $Y \neq 0$. This case returns the group identity. It rejects all other zero-denominator triples.

Lean proves the homogeneous curve equation. It proves that the output is nonzero and projectively nonsingular. It then proves that projective decode equals Mathlib group addition for every valid affine pair. The proof has separate cases for different x-coordinates, doubling, and inverse inputs. The main results are:

- `outputNonsingular`;
- `outputRepresentsSum`;
- `decodeHomogeneous_eq_toAffine`;
- `decodeEvaluateRowSome`;
- `decodeEvaluateRowNone`;
- `perfectCorrectness`.

The axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`.

6 Adaptor optimum

The complete Bosma–Lenstra family has projective parameters $[\alpha : \beta : \gamma]$. Completeness requires $\beta \neq 0$. Normalize $\beta = 1$. Under the exact `Biquadratic` support rule, the adaptor counts are

$$D(0, 0) = 13, \quad D(0, \gamma) = 14 \text{ for } \gamma \neq 0, \quad D(\alpha, \gamma) = 15 \text{ for } \alpha \neq 0.$$

Thus, RCB parameters $[0 : 1 : 0]$ give the unique minimum of 13. This result covers fixed bidegree $(2, 2)$ laws in the current masking architecture.

A second lower bound covers private role-preserving linear rows. A row with t scalar adaptors exposes $14 + t$ selected field values. Its adaptor randomness has dimension at most $2t - 2$. Only the three output coordinates can stay unmasked. Thus, $2t - 2 \geq 11 + t$, so $t \geq 13$. A correct ten-adaptor sharing attempt violates this bound and exposes four extra coefficient invariants. The current row reaches 13. This result does not exclude a nonlinear garbling architecture.

7 Circuit cost

Item	Original 92-row call	Complete 91-row call	Change
Output MAC rows	92	91	−1
Digit adaptors	833	1,188	+355
Bit adaptors	211,582	301,752	+90,170
Fixed-permutation applications	6,322,060	1,508,760	−4,813,300
Public permutation instances	12,495	17,820	+5,325

The first column counts repeated calls in the unoptimized baseline. The last column uses one shared pass for each bit and point row. The public instance count increases because the complete law needs more live support. The Lean direct-table benchmark used 21 measured runs on Apple M5. The complete construction has a 231 ms median. The original baseline has a 989 ms median. The complete construction is 4.28 times faster. It reduces the median by 76.6 percent.

8 Rust row parallelism

The Rust implementation allocates one job for each output row. It allocates the three child nonce domains before Rayon starts. It samples one fresh 32-byte ChaCha12 root from the caller RNG. Row j uses stream

$$0x666d3265635f6d61 + j$$

from word position zero. Indexed collection keeps the row order. Debug builds use the same RNG and nonce schedule in serial order. Release builds use Rayon.

The root seed is erased after row construction. Jobs borrow output keys. They do not copy those secret keys into a second vector. Fixed Boolean arrays replace heap-based coefficient index sets. One byte packs each fixed-key level and digest-pair index. Eight injective 54-byte inputs derive the 15 AES keys. Each input and its padding fit in one SHA-256 compression block. Each 256-bit digest supplies two 128-bit keys; one final half is unused. The 384-bit hash-to-field reducer reads the three AES blocks directly. It avoids generic slice loops and a 48-byte staging copy. The 256-bit field pad uses four explicit little-endian limb XOR operations. Tests compare one-worker and four-worker tables. A second test compares all 273 allocated child nonces with the serial schedule.

On the same Apple M5, the ARMv8 AES benchmarks used 100 Criterion samples. The incomplete 92-row serial baseline estimate was 15.934 ms. The complete 91-row, 10-worker row mean is 2.876 ms. Its 95 percent interval is 2.861 to 2.893 ms. The complete row stage is 5.54 times faster than the serial baseline. Direct field-limb XOR lowered its adjacent row mean by 8.7 percent. Direct hash-block reduction then lowered its adjacent mean by 4.5 percent. The one-block KDF lowered its adjacent mean by a further 8.7 percent. Digest pairing then lowered its adjacent mean by 7.9 percent.

The protocol `encode` mean is 68.190 ms. Its 95 percent interval is 67.963 to 68.581 ms. The comparable baseline is 81.309 ms. The complete implementation reduces this time by 16.1 percent. Its HMAC table has 9,668,659 bytes. This is 2,970,240 bytes larger than the incomplete 91-row table.

9 Security scope

The following invariants stay fixed:

- Every candidate reconstructs the same scalar modulo r .
- Every public circuit has exactly 91 output positions.
- The digit alphabet does not change.
- The input keeps 508 Lamport label positions.
- The fixed-key window loads stay 91, 91, and 72.
- Every support flag is public and independent of the scalar.
- One nonzero factor scales all three homogeneous coordinates.
- Each row uses a separate nonce subtree and RNG stream.
- Worker order cannot change the table or caller RNG state.

The Rust selector uses four fixed 91-round tests and `subtle::Choice`. This gives a fixed iteration count. Lean proves that every tested coordinate fits a signed 128-bit integer. The report does not claim compiler-level constant time.

The Rust checked evaluator rejects a coordinate that is at least the base-field modulus before reduction. It then checks the curve equation. A regression test uses $x = p + 1$ and $y = 2$. Reduction would otherwise map this input to the valid generator $(1, 2)$.

Lean proves that the encoding key contains the 508 Lamport label pairs in x-then-y order. It proves that encoding selects the label for each bit of `affineLamportBits`. The theorem is `Lamport.compatible`.

Lean proves exact joint-tape laws for fresh permutation and random-oracle programming. Each map sends (O, t) to the programmed oracle and the old value $O(x)$. Each map is an involution on a finite sample space. Thus, each map preserves the uniform joint PMF exactly. Programming a fixed output alone does not preserve uniformity. The permutation law also composes over every fixed finite target-tape schedule. Both oracle laws preserve the uniform fiber of fresh oracles that match a fixed query transcript. Lean lifts an exact handler equivalence through every bounded adversary oracle program. Exact replay transports each real trace to the recording handler and preserves its result, query list, count, and final state. A shared-sample coupling gives both traced marginals, equal results and query lists, and related final states. Its structural form composes both adversary phases through an exact related-state bridge with zero result-disagreement mass. The bounded form accepts a bridge that returns an exact result or records a bad event. It keeps both marginals exact by using an independent coupling only after a bad event. Lean proves that every result disagreement implies the bad flag.

It also proves that the final bad mass equals the bridge bad probability. The final coupling theorem bounds the result advantage by that probability. The permutation and hash involutions extend to the complete shared simulator state. Fresh state swaps preserve every shared transcript and label invariant. Each safe ideal-oracle query commutes with the matching full-state swap. Thus, every path-safe bounded adversary program commutes with either swap in exact PMF equality. Lean proves the exact mass of every finite forbidden block set. A set with at most k values has mass at most $k \cdot 2^{-128}$. The two-point collision event therefore has mass at most 2^{-127} . Lean records each reached adversary query and its state. The trace length is at most its indexed query budget. Traced real and ideal games erase exactly to the challenge PMFs. Each gives zero mass to paths above the sum of both adversary budgets. If each step forbids at most k blocks, its collision schedule has mass at most $qk \cdot 2^{-128}$. An exact coupling lemma converts disagreement-event mass to the challenge's real-valued advantage. A graph coupling maps each real trace to one ideal trace and proves both marginals. A final generic theorem reduces the complete adaptive property to this trace transport and its Boolean change event. Lean also bounds this change event by any containing reached-collision event. The gate programmer accepts an ordered selected-gate schedule. It preserves the shared invariant and any prior collision flag. Every reached step is fresh unless the final state records a collision. It counts all selected slots exactly. A false-bit gate adds three fixed-key records, and a true-bit gate adds two. A collision-free schedule adds their sum, which is at most three times the gate count. The final oracle also satisfies every selected target after all later gate programs. Lean instantiates this schedule for one complete 254-bit digit adaptor. It proves the requested weighted field result after collision-free programming. Lean also instantiates one schedule for all 13 active adaptors of one complete RCB row. Its 3,302 selected bit gates return all three requested coordinate values in the same final oracle. It then composes the five membership adaptors with all 91 RCB rows. The complete fixed-key schedule has 1,188 digit adaptors and 301,752 selected bit gates. The linked schedule derives its point-layer input MAC with the exact EncPRF transform. It uses the programmed membership result as the whitening-key input. Fixed-key programming preserves the EncPRF and hash oracles. A satisfied linked schedule equals the full pipeline evaluation and returns all requested values. The programming bridge now retargets each selected curve and RCB coordinate. It puts one field residual in the low bit position and puts zero in every other position. Lean proves that this weighted fold equals the residual. It also proves that retargeting preserves every public table. Collision-free programming returns each requested homogeneous value for all 91 rows.

The complete row repairs the known perfect-correctness failure. The generic bounded bridge is complete. The construction still lacks the distributional hop from real public tables to simulated public tables. It also lacks the final adaptive collision bound for that hop. The worst selected branch uses the three hash slots. Its zero-query numerator is

$$3 \cdot 10692 \cdot 91^2 = 265621356 < 2^{28}.$$

The 100-bit margin is 2,814,100 numerator units. A direct five-slot union bound exceeds 2^{28} . Lean proves the 100-bit arithmetic for each exclusive three-hash or two-pad branch. For a bucket with f false selections and t true selections, Lean proves $3f^2 + 2t^2 \leq 3(f+t)^2$ and $3f + 2t \leq 3(f+t)$. Thus, mixed selected bits stay below the three-hash bucket budget. The semantic reduction must still condition on the selected branch. A companion Kriterion change closes the challenge interface defects. It makes `Kriterion.Solution` a closed dependent bundle. It also defines `Kriterion.Benchmark.garble` at a fixed scalar, security parameter, and seed. The security game now samples the complete finite random tape uniformly. The fixed 256-bit seed selects only the reproducible benchmark tape. This change permits exact uniform permutation and random-oracle proofs. The challenge must still reject this entry until the construction-specific adaptive proof is complete.

10 Reproduction

The Lean proof uses Lean 4.24 and the pinned Mathlib checkout. Run:

```
lake build Proof Tests
lake env lean entry/tests/Correctness.lean
lake env lean AxiomCheck.lean
lake exe bench
```

The Rust validation uses Rust 1.94.0. Run:

```
cargo fmt --all -- --check
cargo clippy --all-targets --all-features -- -D warnings
cargo test --all-features
cargo test --release --all-features
RAYON_NUM_THREADS=10 cargo bench --all-features \
  --bench field_mac_to_ec_mac -- 'field_mac_to_ec_mac/g1/garble'
RAYON_NUM_THREADS=10 cargo bench --all-features \
  --bench protocol_bench -- 'encode'
```

References

- J. Bosma and H. W. Lenstra, “Complete systems of two addition laws for elliptic curves,” *Journal of Number Theory*, 1995.
- J. Renes, C. Costello, and L. Batina, “Complete addition formulas for prime order elliptic curves,” EUROCRYPT 2016.
- RustCrypto, *AES 0.8.4 documentation*, ARMv8 run-time detection section.
- Arkworks, *ark-ec 0.5.0*, GLV scalar decomposition implementation.